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UWE Bristol University of the West of England

Evaluating the Consistency of Explainable AI Methods in Hate Speech Detection on Social Media Platforms

Gonzalo Gamez

Supervised by: Dmitry Pavlyuk

Double degree in Computer Science:

Data Analytics and Artificial Intelligence

TSI and UWE

AI Act

Article 86 of the EU Artificial Intelligence Act (European Union, 2024) establishes the "Right to Explanation of Individual Decision-Making." This legislative development underscores the growing recognition that explainability is not merely a technical consideration but a fundamental requirement for responsible AI deployment.

XAI for Explainability

- Explainable Artificial Intelligence (XAI) is an emerging area of research in the field of Artificial Intelligence (AI). XAI can explain how AI obtained a particular solution (e.g., classification or object detection) and can also answer other 'wh' questions (such as "why", "when", "where", etc.) . This explainability is not possible in traditional AI. (Gohel, Singh and Mohanty, 2021)

Research Object and Subject

- The research object of this study is explainable artificial intelligence (XAI) methods applied to hate speech detection models on social media.
- The research subject is defined as the consistency of XAI methods when applied to AI models that detect hate speech.

Research Question and Hypothesis

- How do LIME and SHAP compare in terms of the consistency of their generated explanations for hate speech detection models under repeated applications?
- The hypothesis is that SHAP provides more consistent explanations than LIME when applied to hate speech detection models, as measured by consistency metrics.

Literature Review

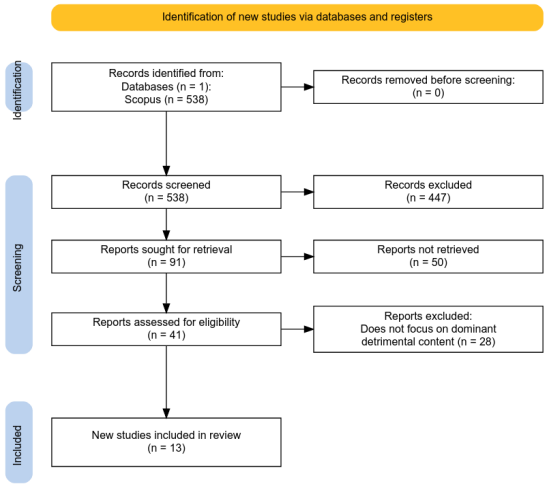


Fig. 1: PRISMA Flow Diagram of Included Studies

We found the following:

- XAI methods - LIME and SHAP
- Deep Learning models – LSTM and BERT variants
- Few used metrics to measure XAI explanations
- Inconsistent selection for which samples to explain - a common method was to select high accuracy predictions of TP, FP, TN, FN

Reference	Dataset Source	Classification Task	XAI Used on AI Model	XAI Method	XAI Use
Ansari, Kaur and Saxena (2023)	YouTube, Facebook, Twitter	Binary	LSTM	LIME, Integrated Gradient	4 examples: Local explanations using and metrics of AOPC, log-odds, and coherence
Hashmi et al. (2024)	Twitter datasets and multilingual corpora	Binary	SVM	LIME	23 examples: Two for each language, except one with 20 examples (1.00 prob.)
Wich et al. (2021)	Twitter	Binary	DistilBERT with GraphSage	SHAP	6 examples: Explanation of each sub model, 1 example with 1.00 prob. Also used masking strategies for component and feature level
Imbwaga, Chittaragi and Koolagudi (2024)	YouTube	Binary	RF	LIME	2 examples: Hate-labelled examples in English and Kiswahili
Nandini and Schmid (2023)	Twitter	Multiclass	BERT	LIME	4 examples: One hate-labelled + aggregated features from 50 examples/class
Siddiqui et al. (2024)	Twitter and cyber bullying datasets	Fine-grained	XLM-RoBERTa	LIME	9 examples: TP, TN, FP, FN + 5 fine-grained samples, 4 with 1.0 prob
Tiwari (2024)	Twitter	Binary	LSTM	LIME	4 examples: One for each LSTM word embedding method
Yadav, Kaushik and McDaid (2023)	Twitter	Binary	NB-B	LIME	4 examples: One TP and one FP, with one at 1.0 prob
Babaeianjelodar et al. (2022)	Twitter	Multiclass	XGBoost	SHAP	1 example: Explained hate-labelled tweet
Hashmi and Sule Yildirim Yayilgan (2024)	Facebook, Twitter, Resset	Multiclass	FAST-RNN	LIME	12 examples: Two per class + two misclassified
Mehta and Passi (2022)	Twitter, Gab, Wikipedia	Binary	RF, NB-G, DT, LR	LIME	4 examples: DT, RF, LR, NB models. IOU F1, Token F1, AUPRC, Comprehensiveness, Subgroup AUC, etc.
Mazhar Qureshi, Qureshi and Rashwan (2023)	Twitter	Multiclass	SVC	LIME, SHAP, Counterfactual	21 examples: SHAP/LIME examples across classes + counterfactual text edits
Hareem Kibriya et al. (2024)	Twitter	Fine-grained	LSTM	LIME, SHAP	7 examples: 4 LIME (weighted features), 2 SHAP-based instances

Table 1: Literature Review of XAI for hate speech detection on social media

Definition of Hate Speech

- *"**any kind of communication** in speech, writing or behaviour, that **attacks** or uses **pejorative** or **discriminatory** language with reference to a person or a group on the basis of **who they are**"* (United Nations, 2024)

Processing and Merging Datasets

Loaded three distinct datasets: HateXplain (Mathew et al., 2021), MLMA (Ousidhoum et al., 2019), and Measuring Hate Speech(Kennedy et al., 2020).

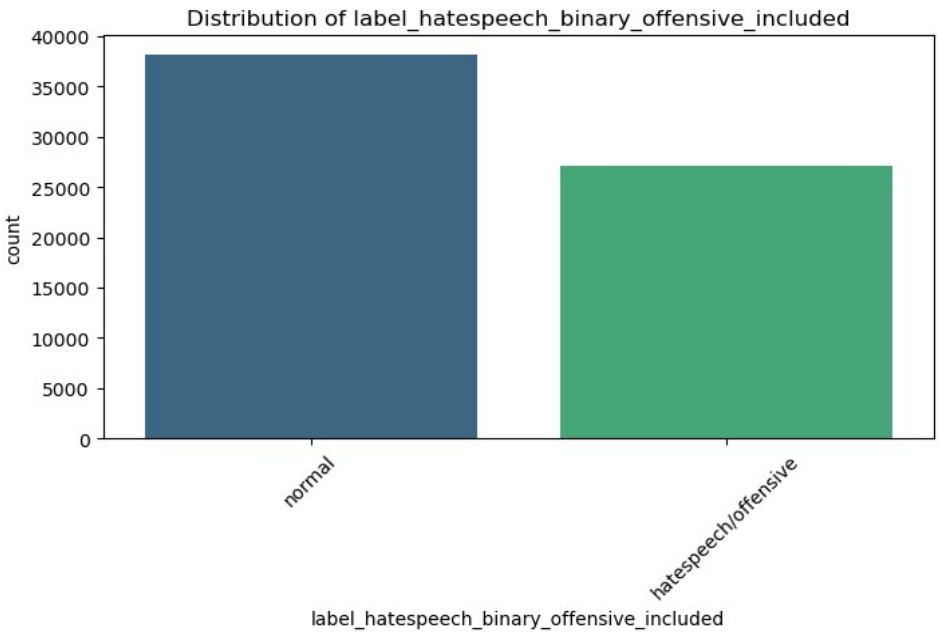


Fig. 2: Distribution of Binary Hate Speech Labels in the Dataset

Methodology Overview

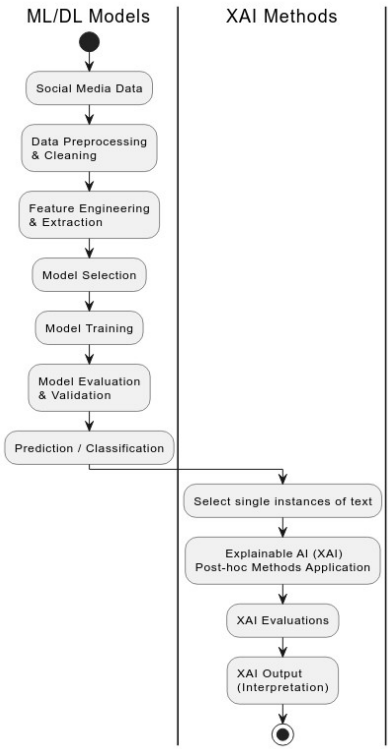


Fig 3: Process for Using XAI Methods on ML/DL models adapted from research from Gongane, Munot and Anuse (2024)

Component	Setting	Value / Description
General	Seeds	4 random seeds between 0 and 100000: [83811, 14593, 3279, 97197] (used for random, numpy, torch, CUDA)
	Model	CardiffNLP (`cardiffnlp/twitter-roberta-base-hate-latest`)
	Dataset Focus	1% of the dataset with `cardiff.score` ≥ 0.95, divided into quarters of True Positives, True Negatives, False Positives, and False Negatives. Sample size per category: 163.
LIME	Explainer Type	`LimeTextExplainer`
	random_state	Same as experimental seeds: [83811, 14593, 3279, 97197]
	Other Settings	Default values used
SHAP	Explainer Type	`shap.Explainer`
	Masker	`shap.maskers.Text` with corresponding model tokenizer
	seed	Same as experimental seeds: [83811, 14593, 3279, 97197]
	Other Settings	Default values used

Table 2: Consolidated Experiment Settings and XAI Parameters

LIME

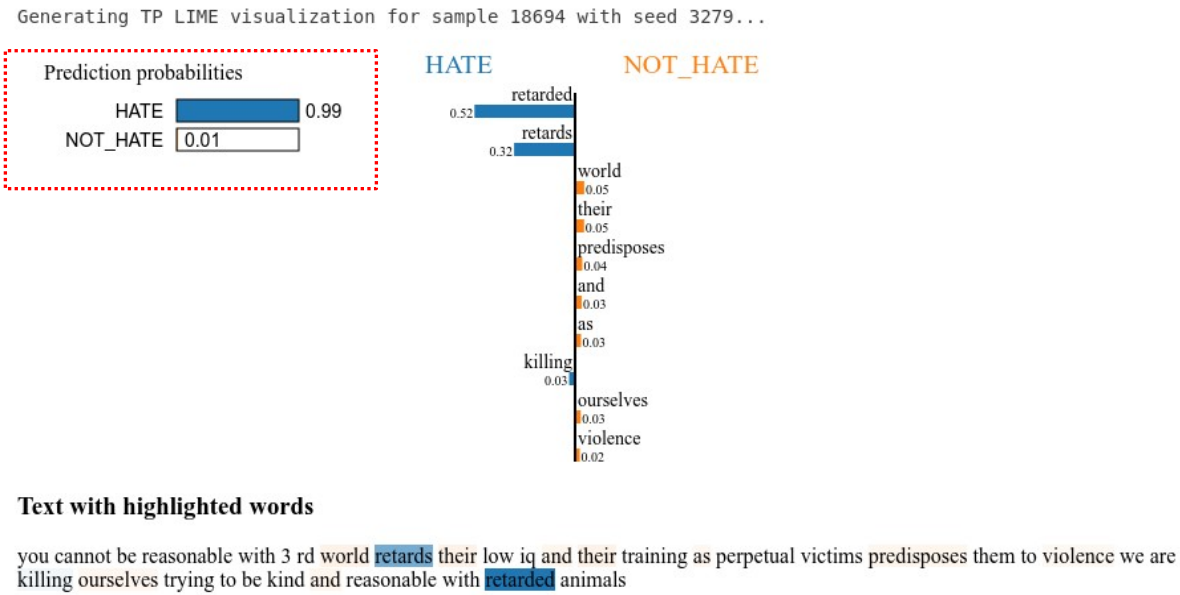


Fig 4: LIME explanation for sample 18694 with seed 3279 TP

id	seed	lime_top5
18694	3279	[('retarded', -0.530), ('retards', -0.296), ('world', 0.050), ('predisposes', 0.042), ('their', 0.041)]
18694	14593	[('retarded', -0.508), ('retards', -0.319), ('their', 0.054), ('killing', -0.041), ('and', 0.038)]
18694	83811	[('retarded', -0.532), ('retards', -0.299), ('their', 0.052), ('predisposes', 0.039), ('world', 0.038)]
18694	97197	[('retarded', -0.515), ('retards', -0.308), ('world', 0.057), ('their', 0.053), ('predisposes', 0.045)]

Table 3: LIME Top-5 Feature Contributions Across Different Seeds

SHAP

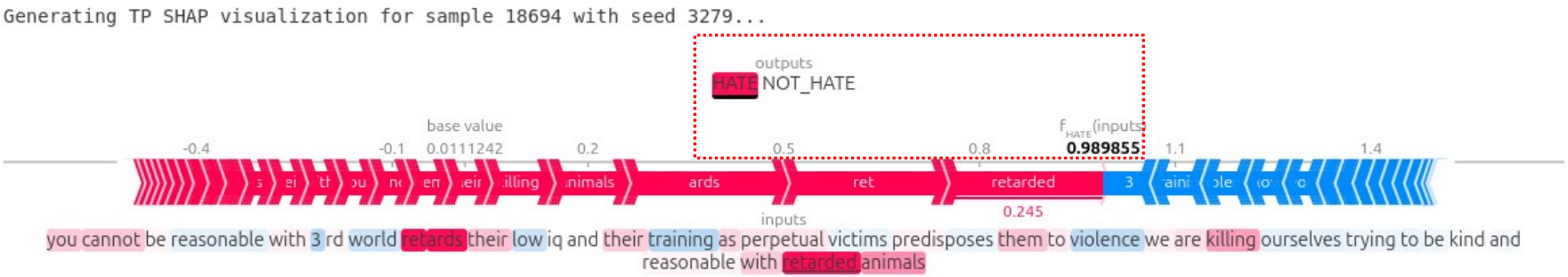


Fig 6: SHAP explanation for sample 18694 with seed 3279 TP

id	seed	shap_top5
18694	3279	[('retarded', -0.245), ('ret', -0.244), ('ards', -0.243), ('animals', -0.116), ('killing', -0.097)]
18694	14593	[('retarded', -0.245), ('ret', -0.244), ('ards', -0.243), ('animals', -0.116), ('killing', -0.097)]
18694	83811	[('retarded', -0.245), ('ret', -0.244), ('ards', -0.243), ('animals', -0.116), ('killing', -0.097)]
18694	97197	[('retarded', -0.245), ('ret', -0.244), ('ards', -0.243), ('animals', -0.116), ('killing', -0.097)]

Table 4: SHAP Top-5 Feature Contributions
Across Different Seeds

Metrics

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|}$$

Jaccard Similarity

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Spearman Correlation

$$\tau_b = \frac{N_c - N_d}{\sqrt{(N_c + N_d + T_X)(N_c + N_d + T_Y)}}$$

Kendall Tau-b

Quantitative Results -LIME

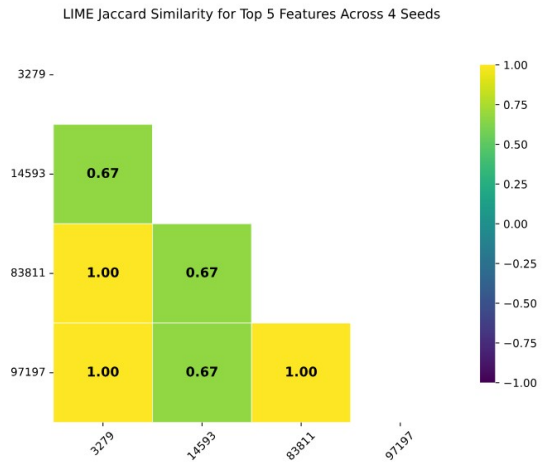


Fig 8. Heat Matrix for Jaccard Similarity across seeds

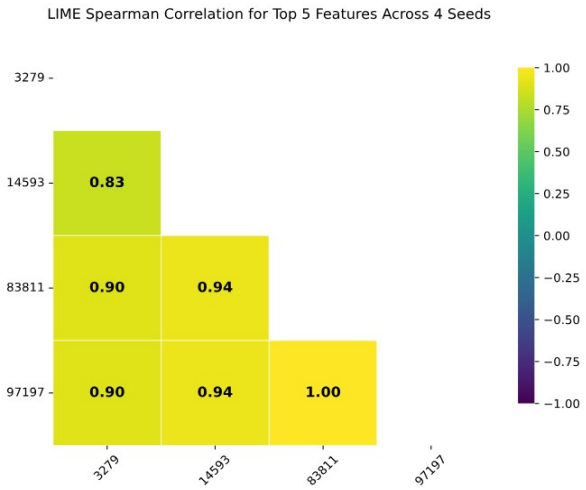


Fig 9. Heat Matrix for Spearman Correlation across seeds

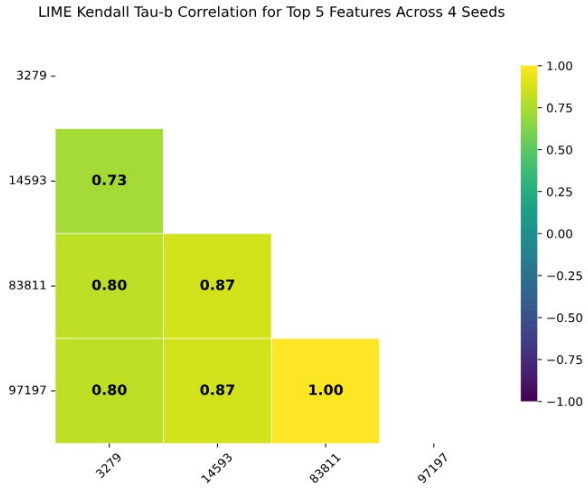


Fig 10. Heat Matrix for Kendall Tau-b across seeds

Quantitative Results -SHAP

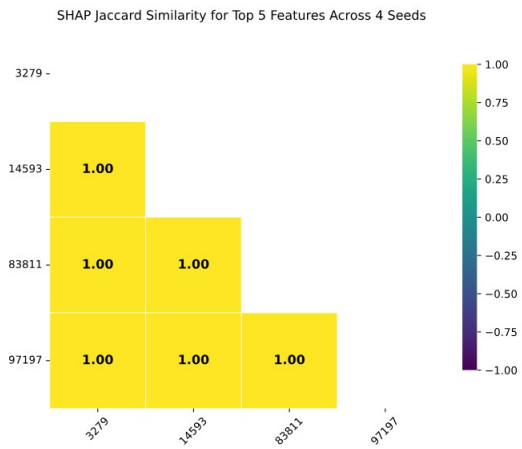


Fig 11. Heat Matrix for Jaccard Similarity across seeds

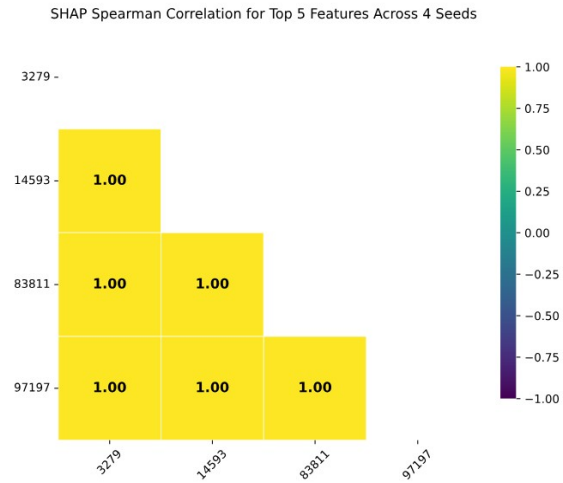


Fig 12. Heat Matrix for Spearman Correlation across seeds

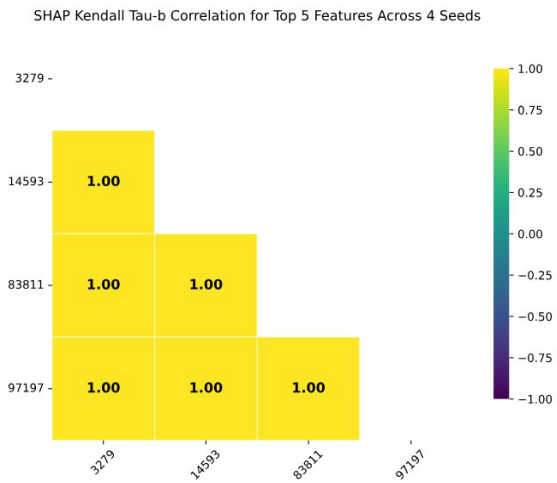


Fig 13. Heat Matrix for Kendall Tau-b across seeds

Aggregated Results over all instances

XAI Method	Jaccard Similarity Mean (std)	Spearman's ρ Mean (std)	Kendall's τ_b Mean (std)
LIME	0.835 (0.193)	0.851 (0.206)	0.776 (0.235)
SHAP	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)

Table 5. Aggregated Consistency Metrics (Mean & Standard Deviation

Key Findings & Implications

- Confirmed hypothesis: SHAP provides more consistent explanations than LIME for hate speech detection
- Practical implications for hate speech text detection systems

Limitations & Future Work

- Dataset limitations and model
- Focus on default parameters only
- Future directions: Other XAI methods, different domains, other detrimental content on social media, parameter tuning effects

Conclusions

- SHAP demonstrates superior consistency for feature contributions provided from explanations compared to LIME across metrics for hate speech text detection

Thank you

Gonzalo Gamez

Supervised by: Dmitry Pavlyuk

St83446@students.tsi.lv

Review

Questions
The thesis highlights SHAP's suitability for content moderation due to its consistency, linking it to user trust and regulatory compliance (EU AI Act, etc.). Question: If you were responsible for integrating an XAI explanation system into a large social media platform for hate speech detection, beyond consistency, what other critical engineering requirements (e.g., latency, scalability, security, cost, explainability for different user personas) would you prioritize, and how would you balance them with consistency?

Acknowledgements

- My Supervisor and Director of the Programme – Professor Dr. sc. Ing. Dmitry Pavlyuk
- Professors During Master's Program
- UWE resources
- State Education Development Agency of the Republic of Latvia for Scholarship

References

All sources at the following link

https://www.mybib.com/j/JudiciousDisgustingTiger

Other XAI metrics

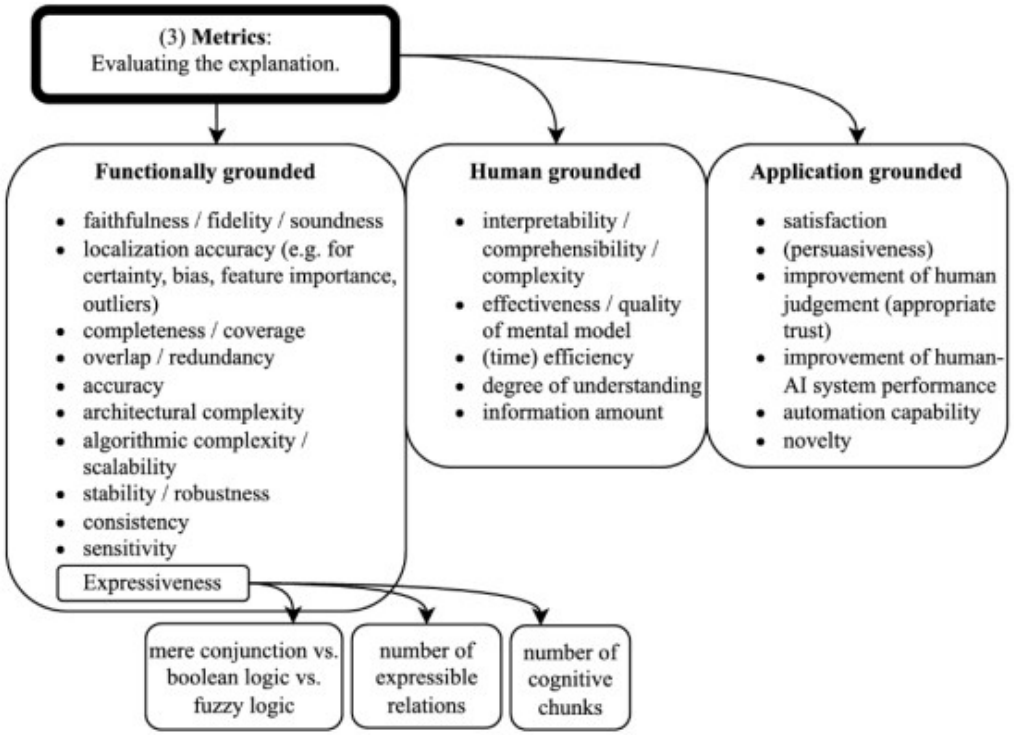


Fig. 7. Overview of XAI metrics by Schwalbe and Finzel (2023)

Extras

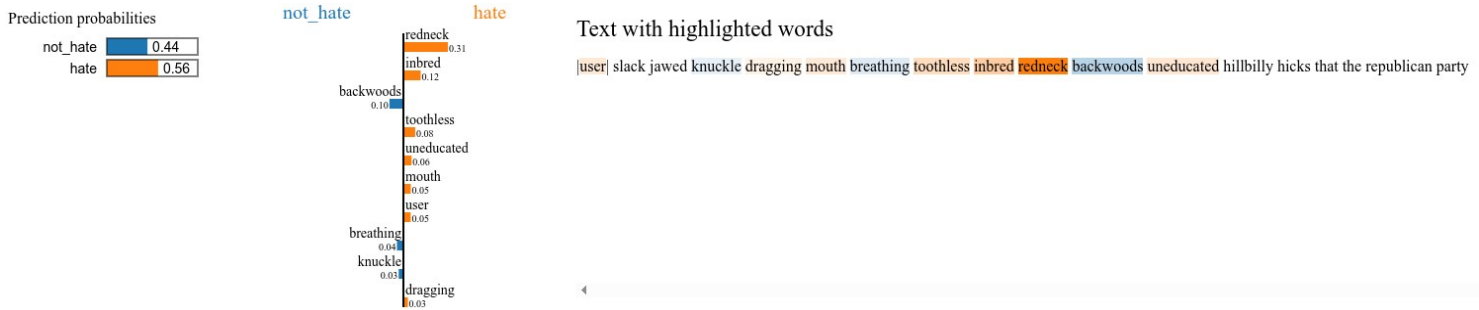


Figure 7: LIME Visualization Showing Word Contributions to a Classification

"Local interpretable model-agnostic explanations (LIME), proposed by Ribeiro, Singh, and Guestrin (2016), is an approach for fitting surrogate models. Surrogate models are trained to approximate the predictions of the underlying black box model." (Molnar, 2022)

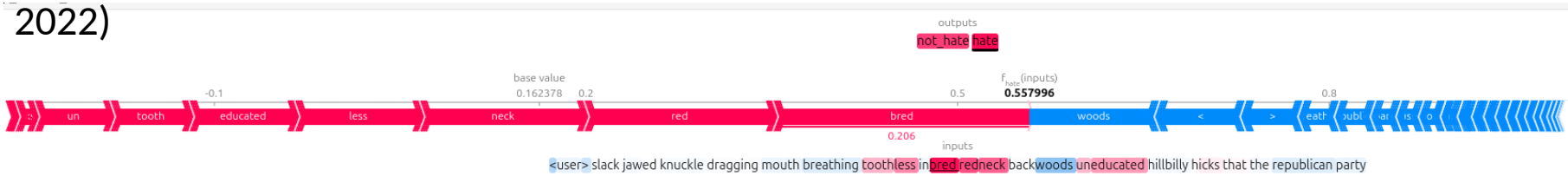


Figure 8: SHAP Visualization of Feature Contributions to a Classification

"SHAP (**SHapley Additive exPlanations**) by Lundberg and Lee (2017) is a method to explain individual predictions. SHAP is based on the game-theoretically optimal Shapley values." (Molnar, 2019)

How LIME works (Molnar,2022)

- **Generate Variations:** LIME creates new text variations by randomly removing words from the original text.
- **Feature Representation:** Each word is represented by a binary feature (1 if included, 0 if removed).
- **Model Prediction:** The model predicts the outcome (e.g., hate speech probability) for each generated variation.
- **Proximity Calculation:** The proximity of each variation to the original text is determined.
- **Local Weight Estimation:** LIME estimates local weights, highlighting words that strongly influence the model's prediction for a specific instance.

How SHAP works (Molnar,2019)

- **Game-Theoretic Approach:** SHAP (SHapley Additive exPlanations) is based on Shapley values from cooperative game theory. It attributes a "payout" (the model's prediction) among features.
- **Feature Contributions:** For text, "features" are typically individual words or tokens. SHAP calculates the contribution of each word to the overall prediction for a specific instance.
- **Coalitions:** It considers all possible "coalitions" (subsets) of words from the text. For each coalition, it evaluates the model's prediction with and without a specific word to determine its marginal contribution.
- **Average Marginal Contribution:** The Shapley value for a word is its average marginal contribution across all possible coalitions, ensuring fair attribution.

LIME (Local Interpretable Model-agnostic Explanations)

Formula: $\xi(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g)$

Variables:

- $\xi(x)$: The explanation for instance x
- g: Interpretable model from class G
- G: Class of interpretable models
- f: Original complex model being explained
- L(f, g, π_x): Measure of how unfaithful g is in approximating f in the locality defined by π_x
- π_x: Proximity measure around instance x
- Ω(g): Complexity measure of the explanation model g

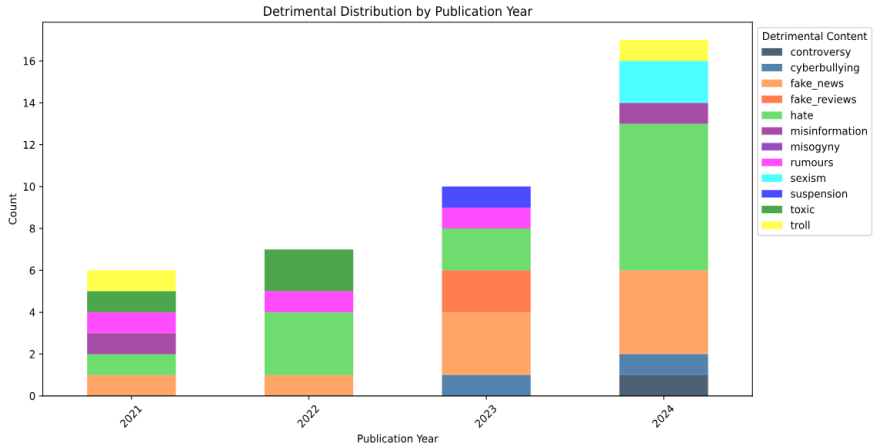
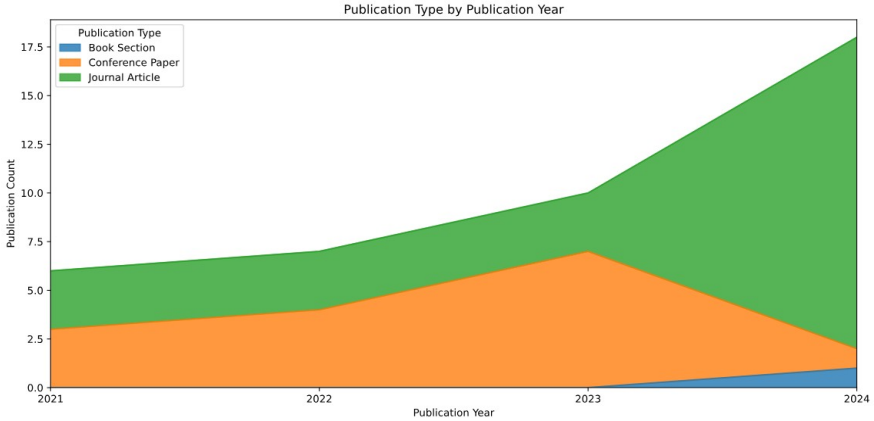
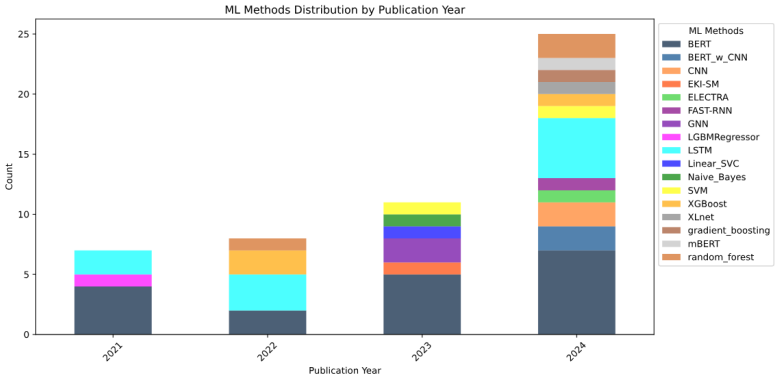
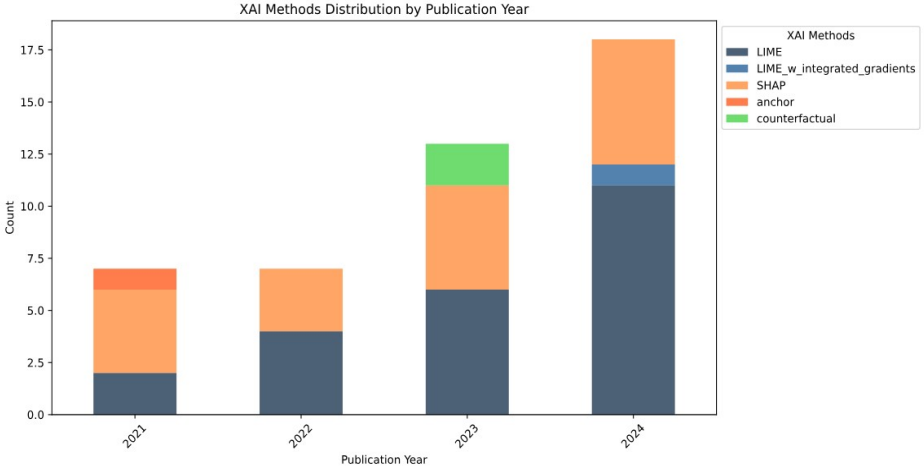
Description: Creates a locally faithful approximation of the original model around a specific prediction using an interpretable model. For text, it typically shows word contributions to the prediction.

SHAP (SHapley Additive exPlanations)

Formula: $\phi_j(f,x) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{j\}) - f_x(S)]$

Variables:

- $\varphi_i(f,x)$: SHAP value for feature i
- f: The model being explained
- x: The specific instance being explained
- F: Set of all features
- S: Subset of features excluding feature i
- |S|: Size of subset S
- |F|: Total number of features
- f(S∪{i}): Prediction with feature i included
- f(S): Prediction with feature i excluded

Description: Based on game theory, assigns each feature an importance value for a particular prediction. The Shapley value represents the average marginal contribution of a feature across all possible coalitions.

Detrimental Content from Search